Solution of Question 1.

(a) i

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

[2]

ii

$$\boldsymbol{G} = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

[2]

iii The syndrome vector is given by

$$\boldsymbol{s}_1 = \boldsymbol{y}_1 \boldsymbol{H}^T = [0\ 1\ 1]\,,$$

hence

$$\hat{\boldsymbol{c}}_1 = [0\ 0\ 1\ 0\ 1\ 1\ 0]\,,$$

and

$$\hat{\boldsymbol{m}}_1 = [0\ 1\ 1\ 0]\,.$$

[3]

iv The syndrome vector is given by

$$[1\ 0\ 1\ 1\ 0\ 0\ 0]\,\boldsymbol{H}^T = [1\ 1\ 0]\,.$$

Let c_4 and c_5 be the missing symbols in $\boldsymbol{c}$. Then one has

$$[c_4\ c_5] \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = [1\ 1\ 0]\,.$$

It is clear that $[c_4\ c_5] = [1\ 0]$. Hence $\boldsymbol{c}_2 = [1\ 0\ 1\ 1\ 0\ 1\ 0]$ and $\boldsymbol{m}_2 = [1\ 0\ 1\ 0]$. [3]

(b) i The distance of a linear code is the minimum number of linearly dependent columns of its parity matrix $\boldsymbol{H}$.
A concise proof is as follows. Any codeword $\boldsymbol{c} = [c_1, c_2, \cdots, c_n] \in \mathcal{C}$

satisfies

$$\sum_i c_i \boldsymbol{h}_i = \mathbf{0},$$

where $\boldsymbol{h}_i$ denotes the i-th column of the parity-check matrix. It is clear that $\mathrm{wt}(\boldsymbol{c})$ gives a number of linearly independent columns of $\boldsymbol{H}$. Hence,

$\min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\mathbf{0}} \mathrm{wt}(\boldsymbol{c}) \geq$ minimum number of linearly dependent columns of $\boldsymbol{H}$.

On the other hand, any set of linearly dependent columns $\boldsymbol{h}_i$ defines a codeword $\boldsymbol{c} \in \mathcal{C}$. Hence $\min_{\boldsymbol{c}\in\mathcal{C}:\ \boldsymbol{c}\neq\mathbf{0}} \mathrm{wt}(\boldsymbol{c}) =$ minimum number of linearly dependent columns of $\boldsymbol{H}$. [3]

ii The distance of the code in Question 1.(a) is 3. Every column is different. Hence every pair of columns are linearly independent and $d(\mathcal{C}) > 2$. There exist three columns that are linearly dependent, for example, columns 1,2,6. Hence the distance is 3. [2]

(c) i

$$x^3 + x^2 + 1 = (x+1)(x^2+1) + x$$
$$x^2 + 1 = x \cdot x + 1.$$

Hence, $f(x) = 1$. [2]

ii

$$\begin{aligned} 1 &= (x^2+1) + x \cdot x \\ &= (x^2+1) + x(x^3 + x^2 + 1 + (x+1)(x^2+1)) \\ &= (x^2+1) + x(x^3 + x^2 + 1) + (x^2 + x)(x^2+1) \\ &= x(x^3 + x^2 + 1) + (x^2 + x + 1)(x^2+1). \end{aligned}$$

Hence, $g(x) = x^2 + x + 1$ and $h(x) = x^3 + x^2 + 1$. [2]

(d) i

$$(z+1)(z^3 + z^2 + z + 1) = z^4 z^1 2 = z^1 6 = z.$$

[2]

ii

$$(z^3+z^2+1)^-1(z^2+1) = z^{-13}z^8 = z^{-5}$$
$$= z^{10} = z^2+z+1.$$

[2]

iii

$$(x-(z^2+z))(x-(z^2+z+1))$$
$$= (x-z^5)(x-z^{10})$$
$$= x^2-(z^5+z^{10})x+z^{15}$$
$$= x^2-(z^2+z+z^2+z+1)x+1$$
$$= x^2+x+1.$$

[2]

Solutions of Question 2.

(a) i Since $p \nmid a$, $\gcd(p, a) = 1$. By Bï¿œzout's Identity, there exist $x, y \in \mathbb{Z}$ s.t. $xp + ya = 1$. Multiply both sides by b, $xpb + yab = b$. Since $p \mid xpb$ and $p \mid ab$, it holds that $p \mid xpb + yab$, i.e., $p \mid b$. [3]

ii The statement is clearly true for $n = 2$ based on the previous sub-question. Suppose that the statement holds for $n = N$ for a given integer $N > 2$. We are going to show that this statement is true for $n = N + 1$.

Let $a = a_1 a_2 \cdots a_N$. Then $p|(aa_{N+1})$. From the previous sub-question, p divides at least one of a and a_{N+1}. Suppose that $p|a$. Then by the hypothesis, $p|a_i$ for some $i \in \{1, 2, \cdots, N\}$. The claimed statement is true. In the other case where $p|a_{N+1}$, the claimed statement also holds. [3]

iii There are infinitely many possible such examples. One of them is as follows. $6 \nmid 2$ and $6 \nmid 3$ but $6|(2 \cdot 3)$. [2]

(b) i As $\gcd(a, b) = 1$, Bï¿œzout's Identity implies that there exist integers x and y such that $1 = ax + by$. Take the modulo arithmetic for both sides. Then $1 = ax \pmod b$. Hence $x = a^{-1} \pmod b$. [3]

ii This can be proved by contradiction. Suppose that $d = \gcd(a, b) > 1$. From the assumption that $a^{-1} \pmod b$ exists, $1 = ax + by$ some integers x, y. As $d = \gcd(a, b)$, it holds that $d|a$, $d|b$, and hence $d|ax + by$. This leads to $d|1$ which contradicts with the hypothesis $d > 1$. [3]

(c) i The complexity of computing the exponential function $g^a \pmod p$ can be as low as $O(\log p)$. Suppose that the binary representation of p has at most $n + 1$ binary digits. Write a into its binary representation $a = d_n d_{n-1} \cdots d_0$ where $d_i \in \{0, 1\}$. Then $g^a \pmod p = (d_n g^{2^n}) \cdot (d_{n-1} g^{2^{n-1}}) \cdots (d_0 g^0) \pmod p$. One can compute $g^2, g^3, \cdots, g^{2^{n-1}}, g^{2^n} \pmod p$ offline and then $g^a \pmod p$ online with complexity $O(\log p)$. [3]

ii The typical way to compute the discrete logarithmic function $\log_g(b)$ is via exhaustive search. This is because the output of $\log_g(b)$ looks random. This is no existing algorithm that computes it efficiently. The computational complexity is $O(p)$ which is infeasible for large p. [2]

iii Let p be a large prime number and g be a primitive element of $\mathbb{F}_p$. The values of p and g are available in public. Alice computes $A = g^a \in \mathbb{F}_p$ and Bob computes $B = g^b \in \mathbb{F}_p$. They exchange these two values. Then

Alice computes $A' = B^a \in \mathbb{F}_p$ and Bob computes $B' = A^b \in \mathbb{F}_p$. It is straightforward to see that $A' = B'$ can serve as the shared secret key. [3]

iv The public key of Alice can be defined as $a = g^{a'} \pmod{p}$.
Bob randomly choose $t \in \mathbb{N}$ and compute $(b^t, ma^t) \in \mathbb{F}_p^* \times \mathbb{F}_p^*$.
To decode the message from Bob, Alice computes $(b^t)^{-a'} \cdot (ma^t) = m \in \mathbb{F}_p^*$. [3]

Solutions of Question 3.

(a) i Let $t = \lfloor \frac{d-1}{2} \rfloor$. Hamming bound states that

$$M \leq \frac{q^n}{\sum_{i=0}^{t} \binom{n}{i}(q-1)^i}.$$

The proof is sketched as follows.
Let $\mathcal{C} = \{\boldsymbol{c}_1, \cdots, \boldsymbol{c}_M\}$. It is straightforward to see that the spheres $S(\boldsymbol{c}_m, t)$ are disjoint. As

$$\bigcup_{m=1}^{M} B(\boldsymbol{c}_m, t) \subseteq \mathbb{F}_q^n,$$

it holds that

$$\sum_{m=1}^{M} \text{vol}\left(B(\boldsymbol{c}_m, t)\right) = M\left(\sum_{i=0}^{t} \binom{n}{i}(q-1)^i\right) \leq q^n.$$

Hamming bound is therefore proved. [4]

ii Singleton bound states $d \leq n - k + 1$. Note for a linear code $\mathcal{C}[n, k]$, its parity check matrix is of the size $(n-k) \times n$. The distance d is the minimum number of linearly dependent columns of $\boldsymbol{H}$. But every subset of $n-k+1$ columns of $\boldsymbol{H}$ is linearly dependent. Hence $d \leq n-k+1$. [2]

(b) i The number of all nonzero column vectors in $\mathbb{F}_2^r$ is $2^r - 1$. The parity check matrix is hence of size $(n-k) \times n = r \times (2^r - 1)$. Hence $n = 2^r - 1$ and $k = 2^r - r - 1$. The distance of a binary Hamming code is clearly 3. Hence a binary Hamming code $\mathcal{H}_r$ has parameters $[2^r - 1, 2^r - r - 1, 3]$. [3]

ii For binary Hamming code $\mathcal{H}_r[2^r - 1, 2^r - r - 1, 3]$, $t = 1$. Hence $\text{vol}(B(\boldsymbol{c}, 1)) = 1 + n = 2^r$. Therefore,

$$\frac{q^n}{\sum_{i=0}^{t} \binom{n}{i}(q-1)^i} = \frac{2^{2^r-1}}{2^r} = 2^{2^r-r-1} = 2^k.$$

Hence, a binary Hamming code is a perfect code. [3]

iii For a binary Hamming code $\mathcal{H}_r$, $n - k + 1 = r + 1$ and $d = 3$. Hence only $\mathcal{H}_2$ is MDS and $\mathcal{H}_r$ is not MDS for all $r > 2$. [3]

(c) i $\forall c(x) \in \mathcal{C}$, write $c(x) = u(x)g(x) + r(x)$ where $\deg(r(x)) < \deg(g(x))$.

As cyclic codes are linear, $r(x) = c(x) - u(x)g(x) \in \mathcal{C}$. At the same time, $g(x) \in \mathcal{C}$ is of the least degree among all nonzero polynomials. It concludes $r(x) = 0$. Hence $c(x) = u(x)g(x)$. [2]

ii In order to construct a BCH code on $\mathbb{F}_2$ with $n = 31$ and $d \geq 7$, consider cyclotomic cosets of 2 modulo 31, including

$$\begin{aligned}
&\mathcal{C}_0 = \{0\}\,,\ \mathcal{C}_1 = \{1, 2, 4, 8, 16\} = \mathcal{C}_2 = \mathcal{C}_4 = \mathcal{C}_8 \\
&\mathcal{C}_3 = \{3, 6, 12, 24, 17\} = \mathcal{C}_6 \\
&\mathcal{C}_5 = \{5, 10, 20, 9, 18\} \\
&\mathcal{C}_7 = \{7, 14, 28, 25, 19\}\,.
\end{aligned}$$

The generator polynomial is given by [3]

$$\begin{aligned}
g(x) &= \mathrm{lcm}(M^{(1)}(x), \cdots, M^{(6)}(x)) \\
&= M^{(1)}(x)M^{(3)}(x)M^{(5)}(x),
\end{aligned}$$

where

$$M^i(x) = \prod_{j \in \mathcal{C}_i} (x - \alpha^j).$$

[2]

iii Let $\boldsymbol{c} \in \mathcal{C}$ and write $\boldsymbol{c} = [c_0, c_1, \cdots, c_{n-1}]$. Since $g(x)|c(x)$, $\alpha, \cdots, \alpha^{n-k}$ are roots of $c(x)$. Hence

$$\boldsymbol{c} \cdot \begin{bmatrix} 1 & \alpha & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{n-k} & \cdots & \alpha^{(n-k)(n-1)} \end{bmatrix}^T = \boldsymbol{c}\boldsymbol{A}^T = \boldsymbol{0}.$$

As every subset of $n-k$ columns of $\boldsymbol{A}$ are linearly independent, $d(\mathcal{C}) \geq n-k+1$. At the same time, by Singleton bound $d(\mathcal{C}) \leq n-k+1$. The only possibility is $d = n - k + 1$. [3]

In fact, this cyclic code is Reed-Solomon code.